

CHAPTER 7: **SOCIAL ISSUES**

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Green Computing
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OBJECTIVES: GREEN COMPUTING

- Consider the following questions:
- What is green computing?
- What are organizations doing to support this initiative?
- What are the pathways to green computing?

GREEN COMPUTING

- **Green computing** is the environmentally responsible and eco-friendly use of computers and their resources.
- In broader terms, it is also defined as the study of designing, manufacturing/engineering, using and disposing of computing devices in a way that reduces their environmental impact.
 - Uses less electricity to run, thus its carbon footprint on the planet is smaller

HOW IT INDUSTRY CAN HARM ENVIRONMENT?

- **Consuming resources** - Non-renewable resources, including precious metals like gold, are used to make technology. Many others, such as coal, are consumed to generate the electricity to use technology. Even some renewable resources, like trees and water, are becoming contaminated or are used up faster than they can renew themselves because of technology.
- **Waste** - Manufacturing technology creates large amounts of waste, and used computers and electronics get thrown out when they break or become outdated. Called "technotrash," these electronics contain all sorts of hazardous materials that are very unsafe for the environment. They need to be disposed of using special methods.
- **Health hazards** - Using toxic materials that can harm our health can cause cancer, and technology addiction can lead to other health problems.

HOW IT INDUSTRY CAN HARM ENVIRONMENT?

- Carbon emissions, mostly carbon dioxide and carbon monoxide, are greenhouse gasses that are produced by people. Greenhouse gasses are gasses in the atmosphere that trap and reflect heat and radiation back to the planet's surface. It is believed that over the last century, the amount of greenhouse gasses in the atmosphere has increased due to carbon emissions, and that they are contributing to global warming.
- Carbon emissions get released into the atmosphere from things like cars, air planes, power plants and factories. They also get released by people like you, when you use a vehicle or electricity created from burning fossil fuels. The computer you're using to read this is using electricity, and so is your mobile device and video game system. We're all guilty of enjoying things that aren't exactly eco-friendly, but if we're smarter about how we use technology, we can reduce our environmental impact.

The amount of carbon dioxide and other carbon compounds emitted due to the consumption of fossil fuels by a particular person, group, etc.

HOW IT INDUSTRY CAN HARM ENVIRONMENT?

- **Technotrash**, also called electronic waste or e-waste, is any broken or unwanted electrical or electronic device, and is currently the most rapidly-growing type of waste.
- If you just throw away technotrash with the regular trash, it usually ends up in a landfill.
- Most electronics contain non-biodegradable materials, and heavy metals and toxic materials like cadmium, lead and mercury. Over time, these toxic materials can leak into the ground, where they can contaminate the water we drink, the plants we eat and the animals that live around the area.
- These toxic materials can cause all kinds of bad effects including nausea, diarrhea, vomiting and even cancer.

THE GOAL OF GREEN COMPUTING

- **Reduces** the use of hazardous materials, **maximise energy efficiency** during the product's lifetime, and to promote recyclability and biodegradability of defunct product and factory usage

GREEN COMPUTING

- To manufacture truly “green” products, companies must:
 - Produce product that requires less electricity
 - Reduce the number of hazardous materials used
 - Increase amount of reusable or recyclable materials
 - Help consumers dispose of their products in an environmentally safe manner at the end of the product’s useful life

GREEN COMPUTING

- Personal computers and cell phones contain thousands of components composed of many different materials
 - Some harmful to humans and environment
 - Beryllium, cadmium, lead, mercury, brominated flame retardants, selenium, polyvinyl chloride
 - Workers along the entire supply chain at risk
 - Users can also be exposed to these materials

GREEN COMPUTING

- **EPEAT (Electronic Product Environmental Assessment Tool)**
 - Enables purchasers to evaluate, compare, and select electronic products
 - Based on a total of 51 environmental criteria
 - Products are ranked in three tiers of environmental performance
- **European Restriction of Hazardous Substances Directive**
 - Restricts use of many hazardous materials in computer manufacturing

GREEN COMPUTING

- How to safely dispose of obsolete computers
 - Many states have recycling programs
 - Some manufacturers have developed programs
 - Greenpeace environmental activist organization
 - Issues quarterly ratings of manufacturers according to the manufacturers' policies on toxic chemicals, recycling, and climate change
- Manufacturers have long way to go to meet the high standards



What are the initiatives in Malaysia?

PATHWAYS TO GREEN COMPUTING

■ Green use

- Intelligent use of energy and information system.
- Reducing the energy consumption of computers and other systems as well as using them in an environmentally sound manner.

■ Green Disposal

- Reduction of waste, reuse and refurbishment of hardware.
- Recycling of out of use peripherals and other items.
- Refurbishment and reuse old computers
- Proper recycling unwanted computer and other electronic equipment

PATHWAYS TO GREEN COMPUTING

■ Green design

- Efficient design of data centers and workstation.
- Design energy-efficient and environmentally sound components, computers, servers, cooling equipment, and data centers.

■ Green manufacturing

- Informed purchasing of components, peripherals, and equipment manufactured with the environment in mind.
- Manufacturing electronic components, computers, and other associated subsystems with minimal impact on the environment.



OBJECTIVE: DIGITAL DIVIDES

- What is digital divides?
- What is being done to reduce the negative influence of the digital divide?

THE DIGITAL DIVIDES

- Standard of living

- Level of material comfort measured by the goods, services, and luxuries available

- Digital divide

- Refers to the **gulf** between some group of people enjoy access to and regularly use the various forms of modern information technology, while others do not
 - Cell phones
 - Personal computers
 - The Internet
- Gulf among age groups, economic classes, and cities/rural areas

THE DIGITAL DIVIDES

- Digital divide must be **bridged** to improve resolution of:
 - Health emergencies
 - Crime emergencies
 - Other emergencies
- Access to IT and communications technology:
 - Enhances learning
 - Provides educational and economic opportunities
 - Influences cultural, social, and political conditions

THE DIGITAL DIVIDES

- Low-cost computers for developing countries
 - One Laptop per Child (OLPC)
 - Provides low-cost laptop computers for education
 - Classmate PC from Intel
 - Eee notebook from Asus
- Mobile phone
 - Tool to bridge the digital divide
 - Costs less than PC and more broadly available

PROGRAM PERANTISISWA KELUARGA MALAYSIA FASA 2

• Diteruskan untuk bantu pelajar B40 di institusi pengajian tinggi awam dan swasta.

• Pelajar B40 daripada keluarga pendapatan isi rumah bawah RM4,850 layak kemuka permohonan.

Diperluaskan kepada pelajar

- 1 Asasi di bawah KPT
- 2 Institut Pendidikan Guru
- 3 Kolej Matrikulasi, Kolej Vokasional & Tingkatan Enam (2 semester) di bawah KPM
- 4 ASWARA di bawah Motac

PERMOHONAN DIBUKA 1 OGOS - 30 SEPT 2022
<https://perantisiswa.kkmm.gov.my>

Agihan Peranti Dijangka pada Oktober / November 2022 bagi fasa 2

SAMSUNG GALAXY TAB A8

YAB DATO' SRI ISMAIL SABRI BIN YAAKOB
PERDANA MENTERI MALAYSIA

OBJECTIVE: WORK ENVIRONMENT

- What are the concern in the workplace related to electronic monitoring?

WORK ENVIRONMENT: MONITORING

- **Monitoring concerns in the workplace**
 - **Blogging:** Blogging represents a possible monitoring risk in the workplace as private thoughts put online can be seen by the anyone. If posting sensitive information and an employer stumbles on to your blog, it is quite possible you could be demoted or fired.

WORK ENVIRONMENT: MONITORING

- **Monitoring concerns in the workplace**
 - **Time Management:** Programs exist that can monitor what webpages you visit on the company's computer. Data analysts can have their performance judged based on how much they type. And when all else fails, video surveillance allows a company to keep records of an employee's activity.

WORK ENVIRONMENT: MONITORING

- **Monitoring concerns in the workplace**
 - **Privacy Concerns:** In addition to being able to monitor what websites you visit, it is also possible and legal for a company to monitor what calls you make from the company phone, and even record the phone call itself.



WORK ENVIRONMENT: MONITORING

- What can be done to protect yourself?
 - If you have a blog, avoid blogging about work.
 - If this isn't desirable, try and privatize entries pertaining to work so that only you or a selected list of people can read them.
 - When blogging about your job, try to be as general as possible - avoid using the company name, names of employers, or specific events or addresses.
 - Also, blogging under a pseudonym can help keep you hidden.



WORK ENVIRONMENT: MONITORING

- What can be done to protect yourself?
 - Be aware of your company's policies on all of these issues. Find out what information they record and how often it is checked.
 - Avoid patently personal use of company time and equipment.



WORK ENVIRONMENT: WEB-SURFING AT WORK

■ Personal Internet Use at Work

- it is not uncommon for everyone to go off on a tangent and surf the web.
- Usage aside from work related material range from sending personal email, online shopping, online gaming, chatting, etc.
- Usage should be based on the employers' policy if one is in place.

OBJECTIVE:

SOCIAL MEDIA NETWORKING

- What are social networks, how do people use them, and what are some of their practical business uses?
- What are some of the key ethical issues associated with the use of social networking Web sites?

WHAT IS A SOCIAL NETWORKING WEB SITE?

- Creates an online community of Internet users that eliminates barriers created by time, distance, and cultural differences
- Allows people to interact with others online by sharing opinions, insights, information, interests, and experiences
- Members may use the site to interact with friends, family members, and colleagues they already know
- Members may also wish to develop new personal and professional relationships

SOCIAL NETWORKING WEB SITE

TABLE 9-1 Popular social networking Web sites

Social networking Web site	Description	Estimated unique monthly visitors
Facebook	Social networking site for keeping up with friends, uploading photos, sharing links and videos, and meeting new people online	700 million
Twitter	A real-time information service for friends, family members, and coworkers looking to stay connected through the exchange of messages that are a maximum of 140 characters	200 million
LinkedIn	Business-oriented social networking site used for professional networking; users create a network made up of people they know and trust in business.	100 million
MySpace	General social networking Web site used by teenagers and adults worldwide; allows members to communicate with friends via personal profiles, blogs, and groups, as well as to post photos, music, and videos to their personal pages	80.5 million

(Continued)



Social networking Web site	Description	Estimated unique monthly visitors
Ning	Platform that enables users to create their own social network following a simple process to name the network, choose a color scheme, and allow for unique profile questions; serves as a portal to access tens of thousands of user-created social networks	60 million
Tagged	Social network with a focus on helping members meet new people; suggests new friends based on shared interests; allows members to browse people, share tags and virtual gifts, and play games	25 million
Google Plus ¹³	Social network operated by Google that integrates social services such as Google Profiles and Google Buzz, and introduces new services such as Circles (enables users to organize contacts into groups for sharing), Hangouts (URLs used to facilitate group video chat), Sparks (enables users to identify topics in which they are interested), and Huddles (allows instant messaging within Circles)	25 million

Source Line: "Top 15 Most Popular Social Networking Websites/July 2011," eBiz/MBA, www.ebizmba.com/articles/social-networking-websites. "Google Plus Reaches 25 Million Users, Activity Declines," Search Engine Journal, © August 3, 2011, www.searchenginejournal.com/google-plus-reaches-25-million-users-activity-declines/31500.

BUSINESS APPLICATIONS OF ONLINE SOCIAL NETWORKING

- **Social network advertising**
 - Uses social networks to communicate and promote the benefits of products and services
- **Social network advertising strategies**
 - **Direct advertising**
 - Banner ads on social networking Web site
 - **Advertising using an individual's network of friends**
 - People frequently make decisions based on input from their close group of friends
 - Ethical issues with exploiting an individual's personal relationships for the financial benefit of a company



BUSINESS APPLICATIONS OF ONLINE SOCIAL NETWORKING (CONT'D.)

- **Social network advertising strategies (cont'd.)**
 - **Indirect advertising through groups**
 - Interested users can join by becoming “fans”
 - Fans gained in this manner may not remain loyal
 - **Company-owned social networking Web site**
 - Users can talk about what new products, services, or improvements they would like to see
 - **Viral marketing**
 - Users pass along marketing message to others, creating the potential for exponential growth



THE USE OF SOCIAL NETWORKS IN THE HIRING PROCESS

- recruiters use some form of social media in the recruiting process
- Employers can and do look at the social networking profiles of job candidates when hiring
- Companies may reject candidates who post:
 - Information about their drinking or drug use
 - Provocative or inappropriate photos
 - Discriminatory remarks relating to race, gender, or religion
 - Confidential information



USE OF SOCIAL MEDIA TO IMPROVE **CUSTOMER SERVICE**

- Consumers use social networks to share their experiences, both good and bad, with others
- Also seek help and advice on how to use products more effectively and how to deal with special situations
- Unless organizations monitor social networks, customers are left to resolve questions and issues on their own, risking loss of customers and future sales



SOCIAL SHOPPING WEB SITES

- Combine two highly popular online activities: shopping and social networking
- Shoppers and sellers can share information and make recommendations while shopping online
- Revenue is generated through retailer advertising or by sharing with retailers data about their members' likes and dislikes
- Retailers can design product improvements based on input and get ideas for new product lines
- Great way for small businesses to boost sales

TABLE 9-3 Sample of social shopping Web sites

Social shopping site	Description
Buzzillions	Product review Web site with over 15 million reviews across a wide range of products, with product rankings based on feedback from customers
Crowdstorm	Price comparison shopping resource that aggregates product information from various online buyers guides, reviews, and blog postings
JustBoughtIT!	Facebook and Twitter app for capturing product recommendations from the online community; users can post a photo or screenshot online, share their purchases, and comment on what others are buying.
Kaboodle	Social shopping site where members can discover, recommend and share new products, provide advice, share feedback, get discounts, and locate bargains
MyDeco	Site with a focus on interior design and home décor; users can mock up virtual rooms using their favorite products
OSOYOU	UK-based social shopping site for women with an interest in fashion and beauty products

Source Line: Course Technology/Cengage Learning.

SOCIAL NETWORKING ETHICAL ISSUES

- Ethical issues for social networking Web sites are:
 - Cyberbullying
 - Cyberstalking
 - Sexual predators
 - Uploading inappropriate material
- Cyberbullying
 - Harassment, torment, humiliation, or threatening of one minor by another minor or group of minors via the Internet or cell phone
 - Cyberbullying can become so intense, child commits suicide



SOCIAL NETWORKING ETHICAL ISSUES (CONT'D.)

- **Numerous forms of cyberbullying**
 - Sending mean-spirited or threatening messages
 - Sending thousands of text messages to victim's cell phone and running up a huge cell phone bill
 - Impersonating victim and sending inappropriate messages to others
 - Stealing victim's password and modifying his or her profile to include racist, homophobic, sexual, or other inappropriate data that offends others or attracts the attention of undesirable people



SOCIAL NETWORKING ETHICAL ISSUES (CONT'D.)

- Numerous forms of cyberbullying (cont'd.)
 - Posting mean, personal, or false information about the victim in the cyberbully's blog
 - Creating a Web site whose purpose is to humiliate or threaten the victim
 - Taking inappropriate photos of the victim and either posting online or sending to others via cell phone
 - Setting up an Internet poll to elicit responses to embarrassing questions regarding victim
 - Sending inappropriate messages while playing interactive games

SOCIAL NETWORKING ETHICAL ISSUES (CONT'D.)

■ Cyberstalking

- Threatening behavior or unwanted advances using the Internet or online and electronic communications
- Adult version of cyberbullying
- Can escalate into:
 - Abusive or excessive phone calls
 - Threatening or obscene mail
 - Trespassing
 - Vandalism
 - Physical stalking
 - Physical assault

SOCIAL NETWORKING ETHICAL ISSUES (CONT'D.)

- **Uploading of inappropriate material**
 - Social networking Web sites have policies against uploading videos depicting violence or obscenity
 - Most social networking Web sites have terms of use agreements that give the sites the right to delete material and terminate users accounts that violate their policy
 - Most Web sites do not have sufficient resources to review all material posted



EVALUATING INFORMATION

- Does the openness of the web increase distribution of useful or inaccurate, foolish, and biased information?

